Introduction to Theoretical Computer Science

CS2490 - Session 9: DFA minimization

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Fall 2026

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Recall from last session

Any NFA $\mathcal{A} = (\Sigma, Q, \Delta, I, F)$ can be turned into a DFA $\mathcal{A}^D = (\Sigma, Q^D, \delta^D, q_0^D, F^D)$.

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- then add new subsets as you go: if $\delta^D(a, S)$ is undefined for some $S \in Q^D$ and $a \in \Sigma$, define $\delta^D(a, S) = \bigcup_{q \in S} \Delta(a, q)$ and add it to Q^D,

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- until no new subset can be added.

Finally, $q_0^D = I$ and $F^D = \{S \in Q^D \mid S \cap F \neq \emptyset\}$.

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- If there exists a NFA $\mathcal{A}$ recognizing L, then there exists a DFA $\mathcal{A}^D$ recognizing L (by the subset construction). Since $\mathrm{L}(\mathcal{A}^D) = \mathrm{L}(\mathcal{A}) = L$, the language L is regular.

□

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First option: manual proof that every word accepted by $\mathcal{A}$ is also accepted by $\mathcal{B}$, and vice versa.

Second option: transform each automata into a "canonical" form, and check if the two canonical forms are identical.

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Main idea: eliminate the unreachable states and merge together states that do "the same job".

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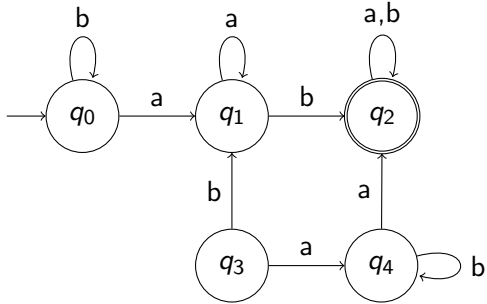
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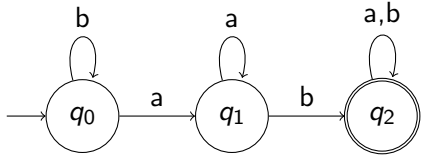
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Important: Removing unreachable states gives a trimmed DFA equivalent to $\mathcal{A}$.

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In other words, p and q are distinguishable if and only if you can read from p a word leading to acceptance that does not lead to acceptance when read from q (or vice versa).

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$\mathcal{A}^{min}$ is called the minimization of $\mathcal{A}$.

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Proof: $w \in L_{\delta(a,p)} \iff \delta^*(w, \delta(a, p)) \in F \iff \delta^*(aw, p) \in F \iff aw \in L_p \iff aw \in L_q \iff \delta^*(aw, q) \in F \iff \delta^*(w, \delta(a, q)) \in F \iff w \in L_{\delta(a,q)}$.

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Proof: by induction on $|w|$. Base case $w = \epsilon$ is trivial. For $w = av$ with $a \in \Sigma$ and $v \in \Sigma^*$, we have:

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$\implies$ $\mathrm{L}(\mathcal{A}^{\min}) = \mathrm{L}(\mathcal{A})$.

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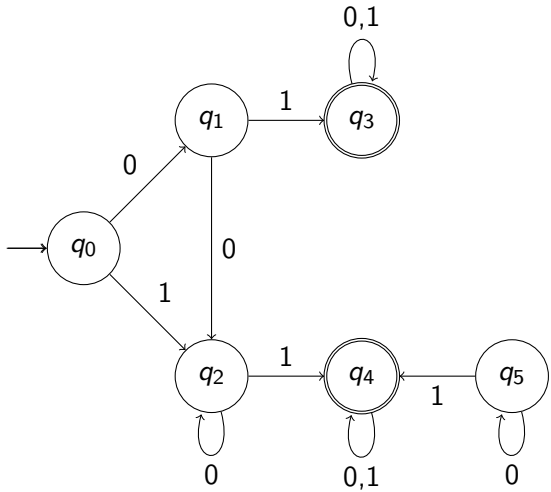
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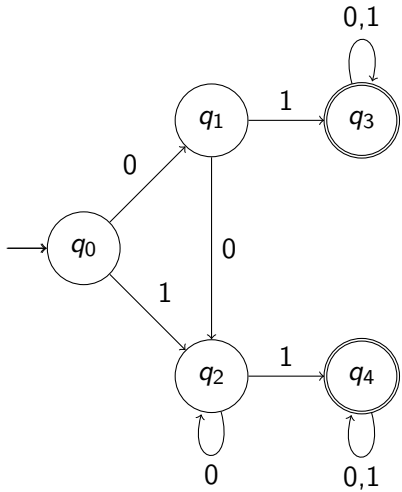
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Proof: for each state χ of $\mathcal{A}^{\min}$, Claim 3 gives us a word w_χ such that $\delta^{\min *}(w_\chi, {q_0}^{\min}) = \chi$. For any DFA $\mathcal{B}$ equivalent to $\mathcal{A}$, write q_χ for the state reached by w_χ in $\mathcal{B}$. All the states q_χ must be distinct: if $q_\chi = q_{\chi'}$ for some $\chi \neq \chi'$, then $w_\chi u$ and $w_{\chi'} u$ would be accepted or rejected together by $\mathcal{B}$ (and thus by $\mathcal{A}^{\min}$) for any $u \in \Sigma^*$, preventing us to find a witness of distinguishabilty for χ and χ'. Thus, $\mathcal{B}$ has at least as many states as $\mathcal{A}^{\min}$.

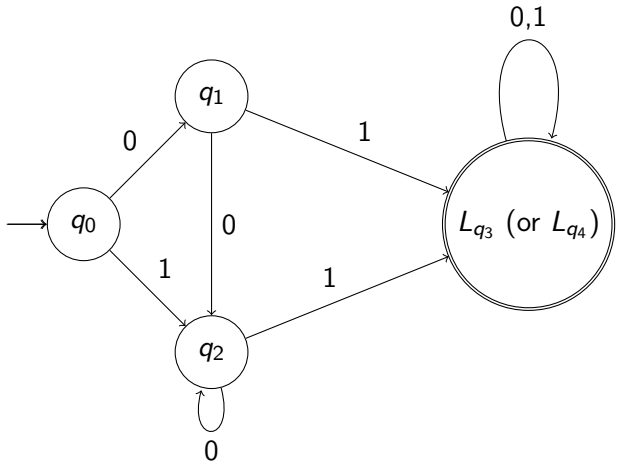
Example of minimization



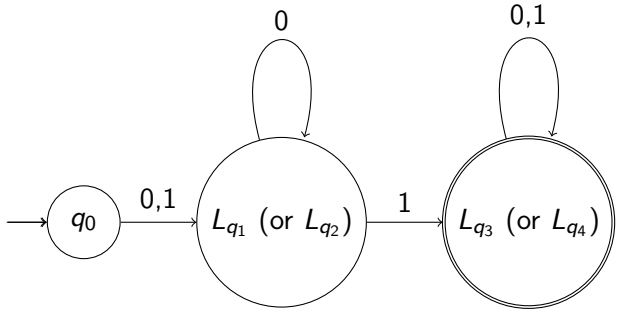
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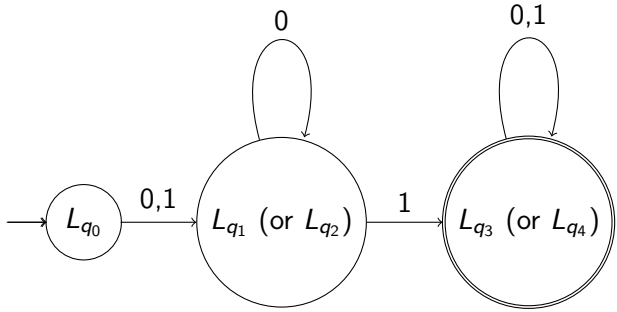
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4. Repeat until no new marking: for each unmarked pair (p, q), if there exists $a \in \Sigma$ such that $(\delta(p, a), \delta(q, a))$ is marked with k, then mark (p, q) with $k + 1$.

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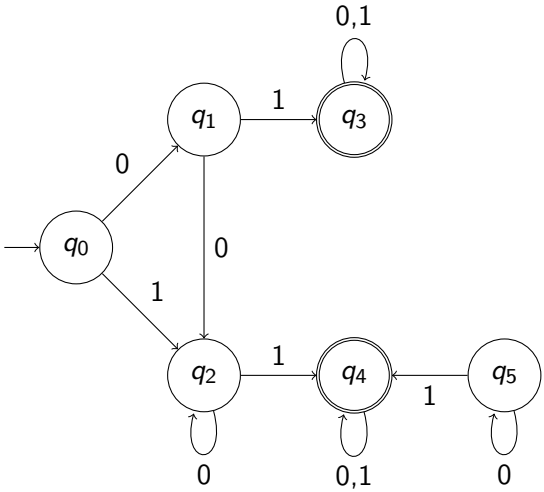
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6. Define the transitions, the initial state and the accepting states accordingly.

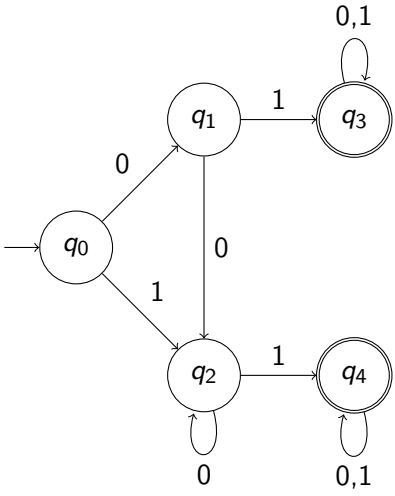
Algorithm: example



	q_0	q_1	q_2	q_3	q_4
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q_1					
q_2					
q_3					
q_4					

- Step 1: Trim and list all pairs.

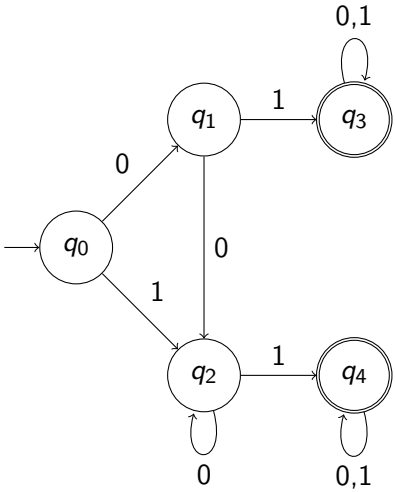
Algorithm: example



	q_0	q_1	q_2	q_3	q_4
q_0				0	0
q_1				0	0
q_2				0	0
q_3	0	0	0		
q_4	0	0	0		

- Step 2: Mark pairs (p, q) where one is accepting and the other is not (0).

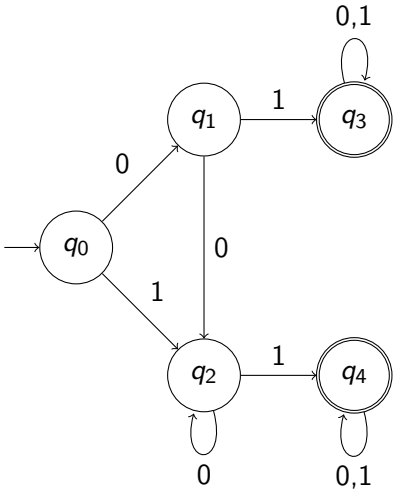
Algorithm: example



	q_0	q_1	q_2	q_3	q_4
q_0		1	1	0	0
q_1	1			0	0
q_2	1			0	0
q_3	0	0	0		
q_4	0	0	0		

- Step 3: Propagate markings (1) if transitions lead to marked pairs.

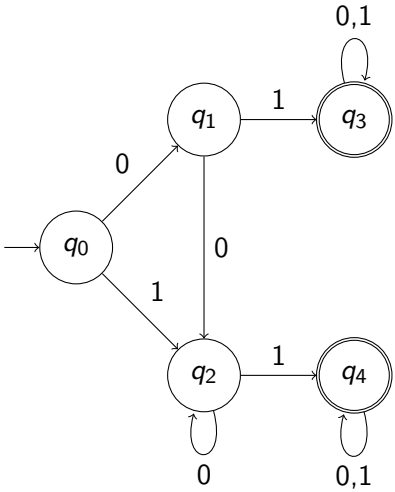
Algorithm: example



	q_0	q_1	q_2	q_3	q_4
q_0		1	1	0	0
q_1	1			0	0
q_2	1			0	0
q_3	0	0	0		
q_4	0	0	0		

- Step 4: Continue propagating until no new markings.

Algorithm: example



	q_0	q_1	q_2	q_3	q_4
q_0	✗	1	1	0	0
q_1	1	✗	✓	0	0
q_2	1	✓	✗	0	0
q_3	0	0	0	✗	✓
q_4	0	0	0	✓	✗

- Step 5: Unmarked pairs are indistinguishable, merge them.